

MEGN 441

Introduction to Robotics

Zhaoda Du

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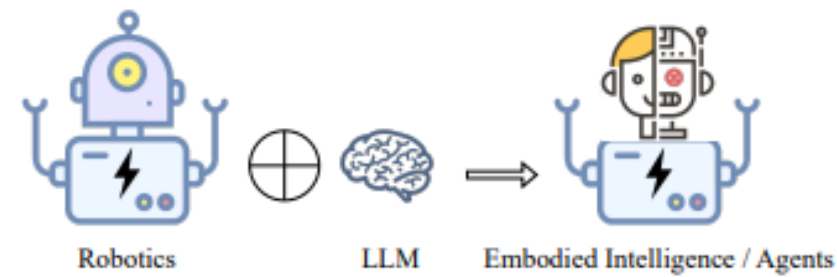


Who am I?

A PhD student.

My research:

1. LLM+Robotics



2. Assistive Technology for Blind user

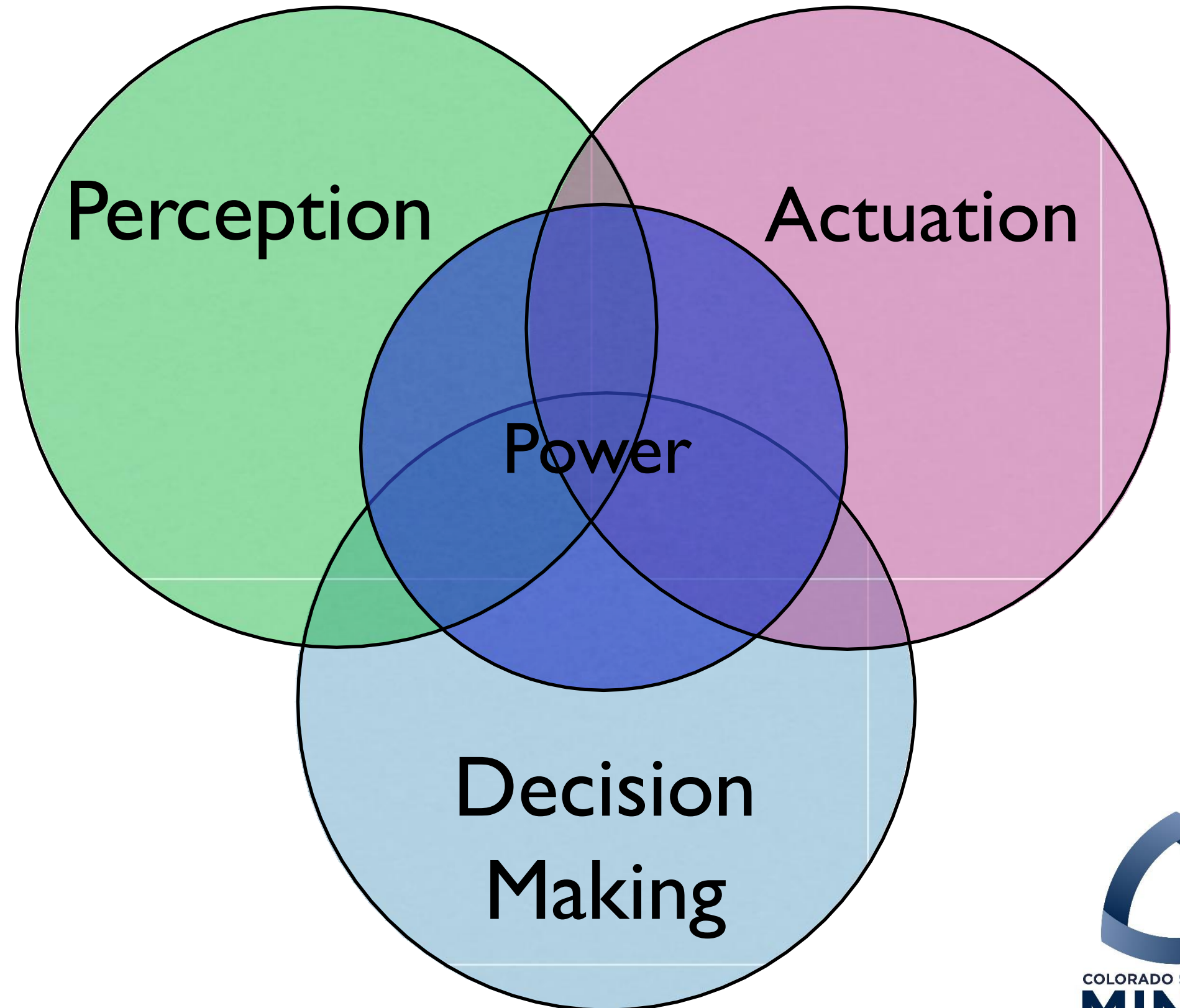


Now imagine you are blind,
standing in the cooking area.

What is a robot?

Major Robot Subsystems

A robot is a machine—often **programmable**—that can sense its environment, process information, and **act on that environment** to achieve a goal, **either autonomously or under human control**.

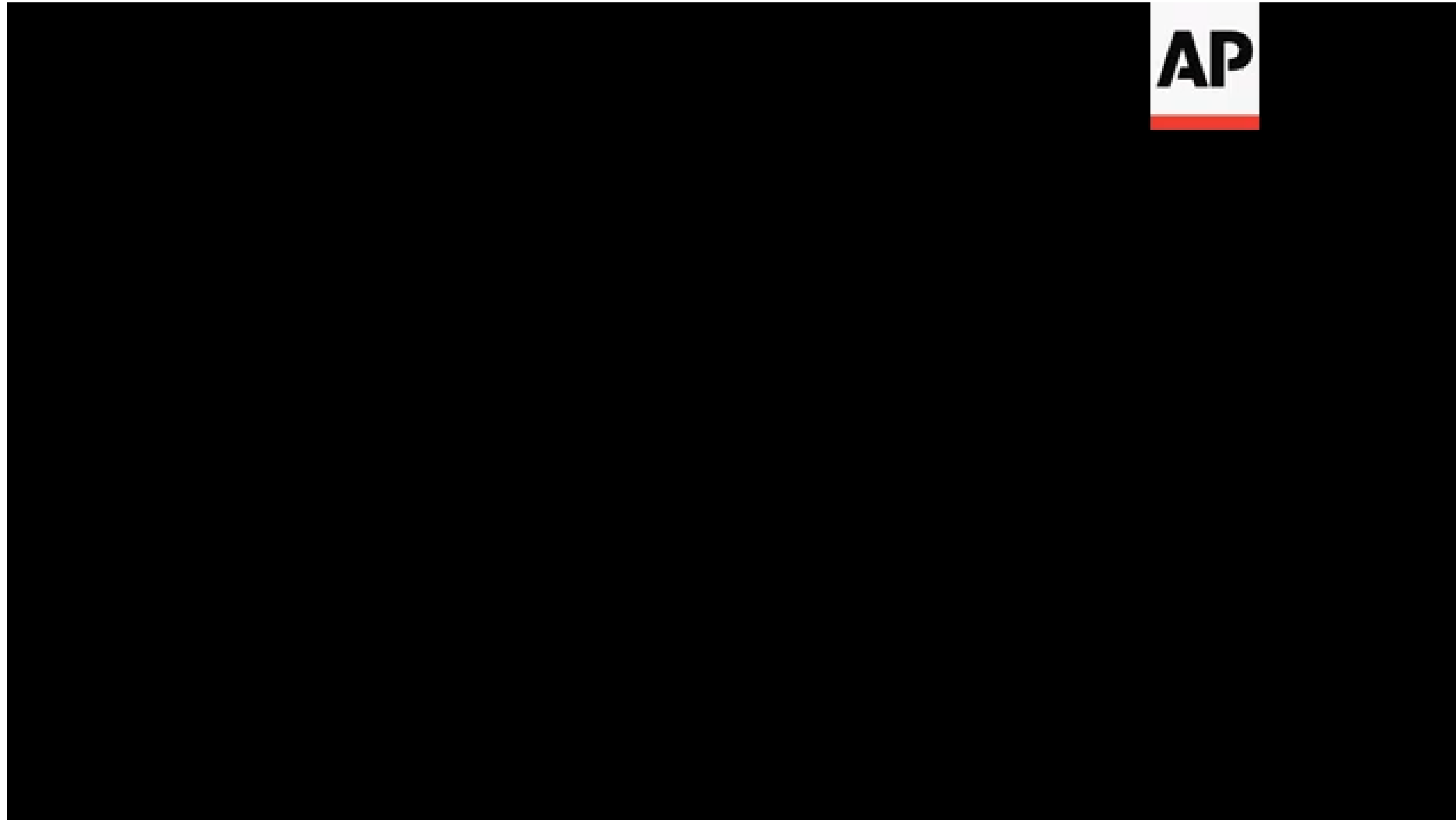


Anatomy of a Robot

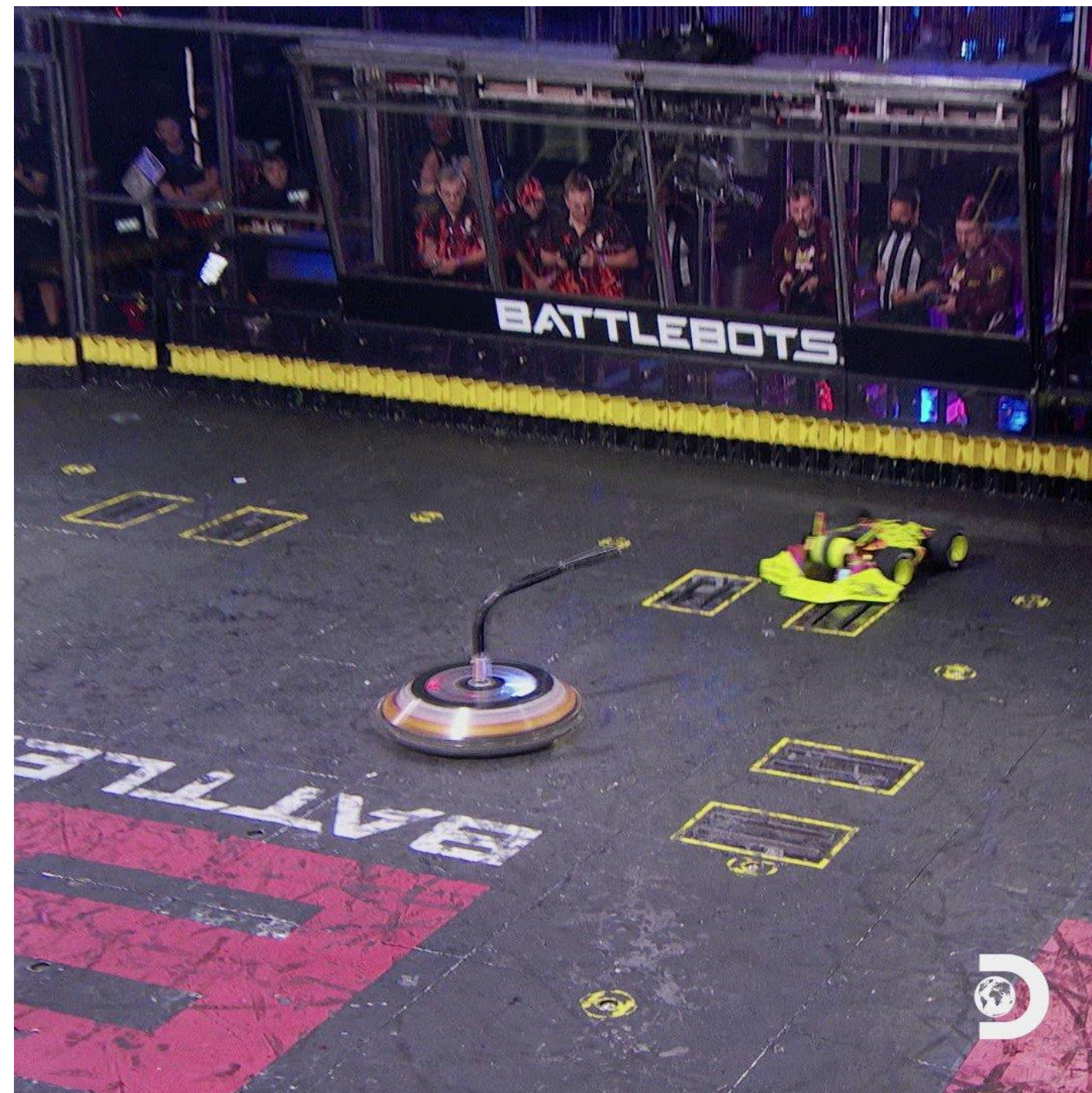
- **Sensing (Perception):** Robots use sensors (e.g., cameras, microphones, touch, GPS, lidar) to gather information about their surroundings.
- **Computation (Processing/Decision-Making):** A robot has a control system—this can be as simple as a circuit or as advanced as AI—that interprets sensor data and decides what to do.
- **Action (Actuation):** Robots use actuators (motors, wheels, arms, grippers, etc.) to physically interact with and change their environment.

What year was the first programmable robot built?

Unimate (1956, video from 1961)



Robot... or not?



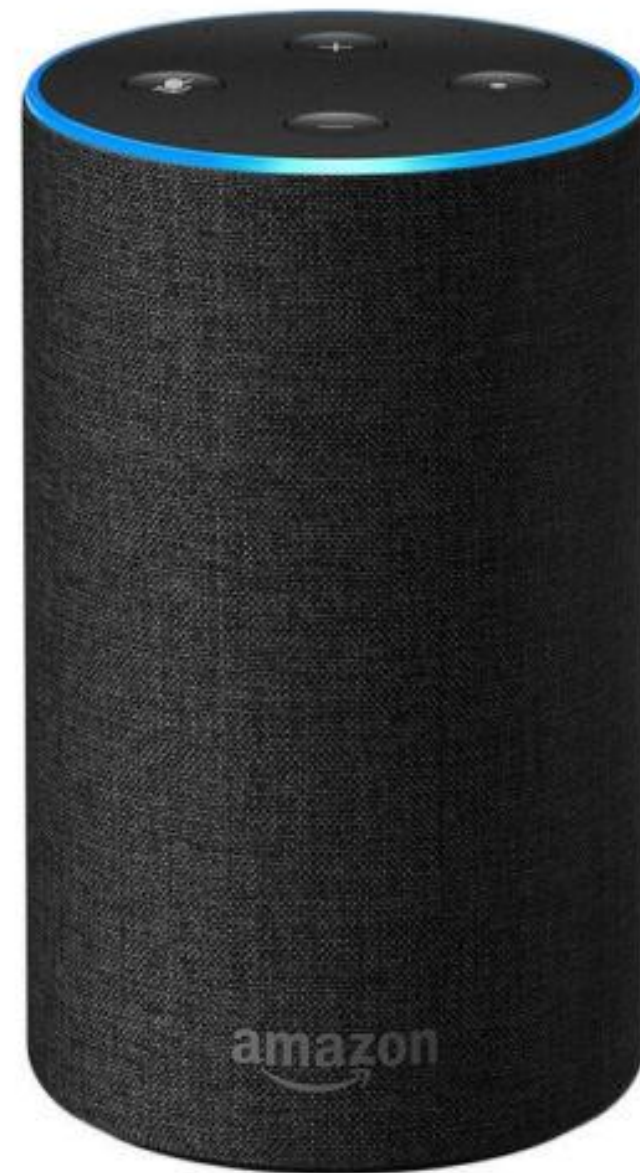
teleoperated robots

Robot... or not?



exoskeleton robot

Robot... or not?

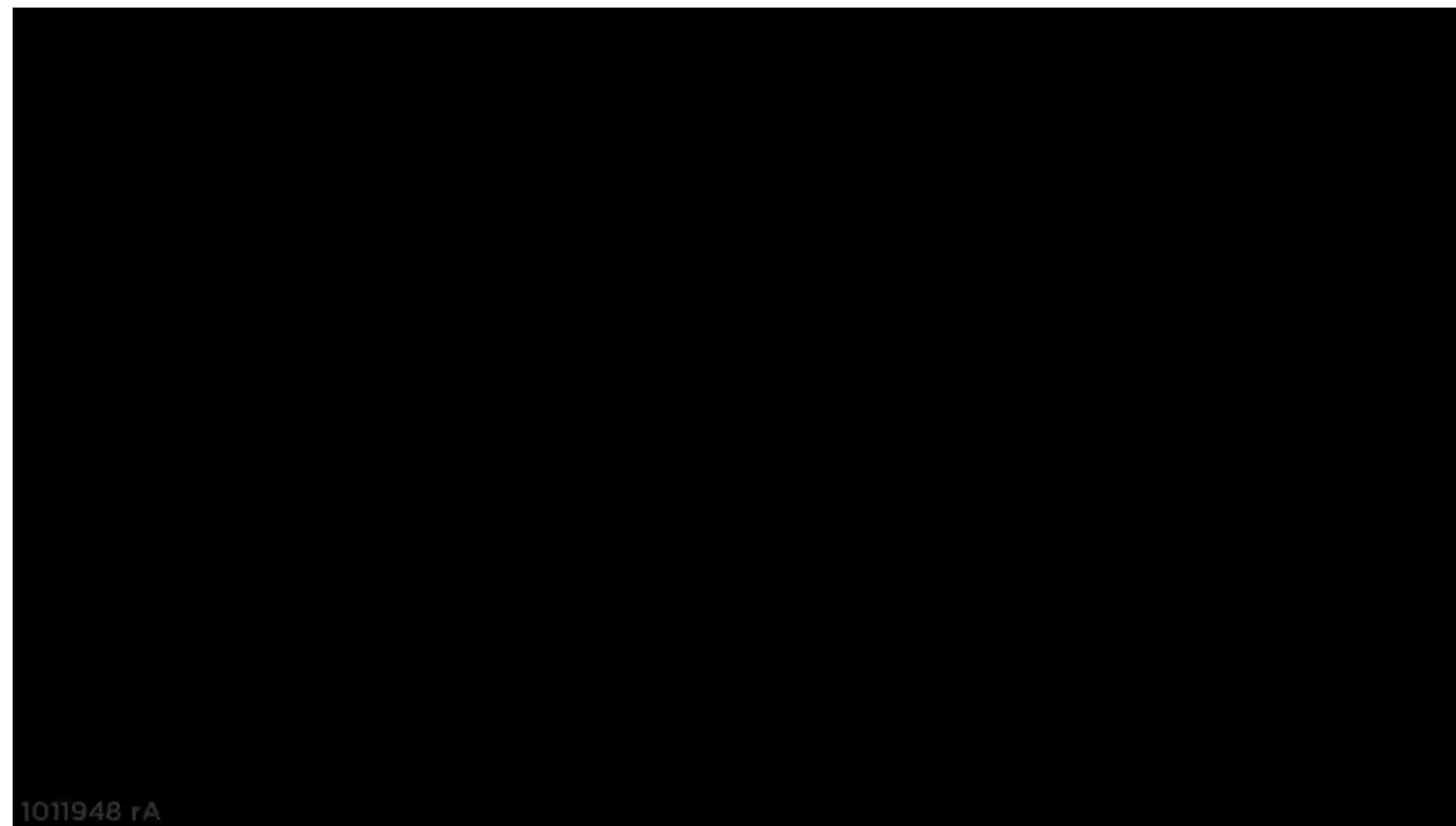


Robots Applications

- Industry & Manufacturing Robots



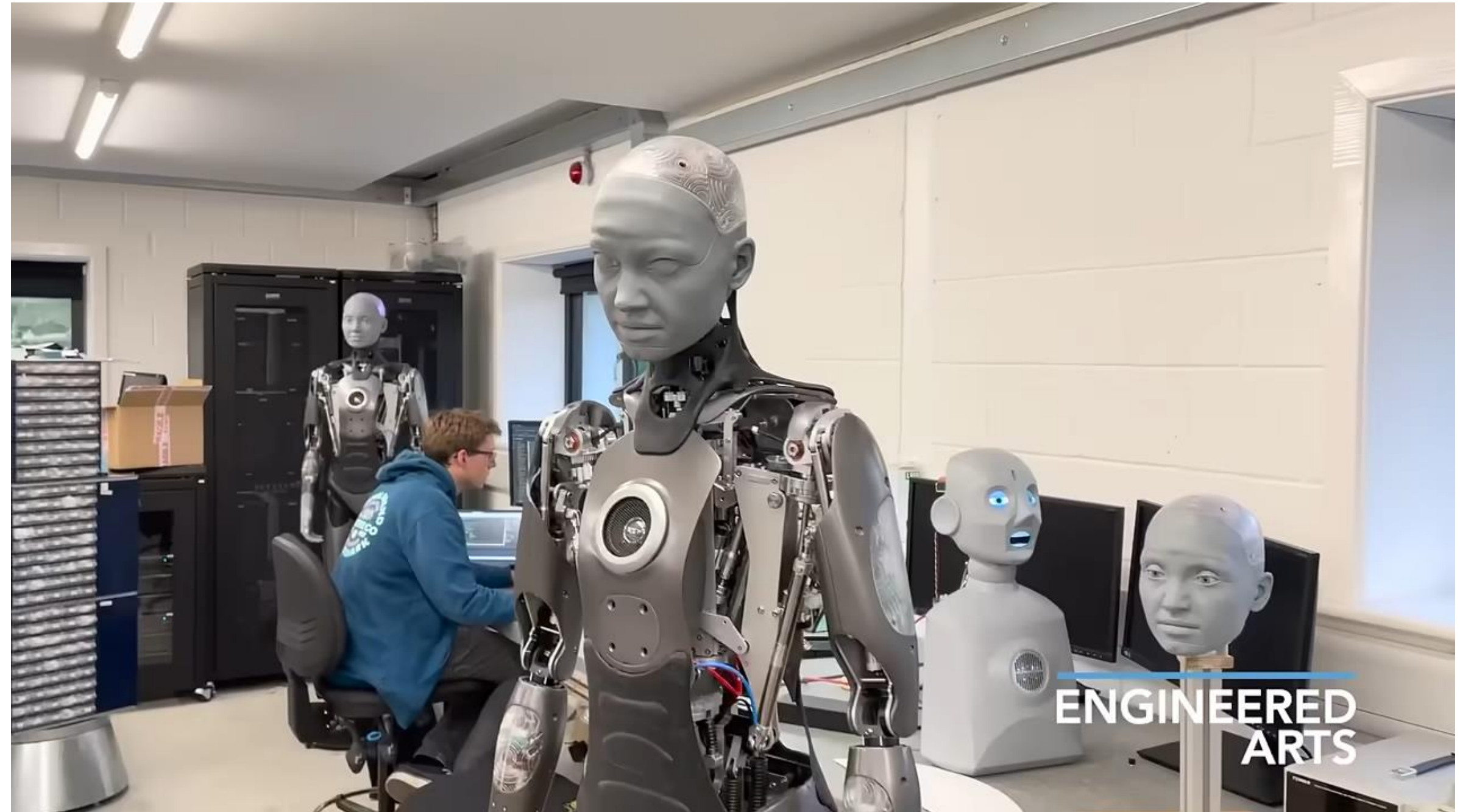
Healthcare



Home & Daily Life

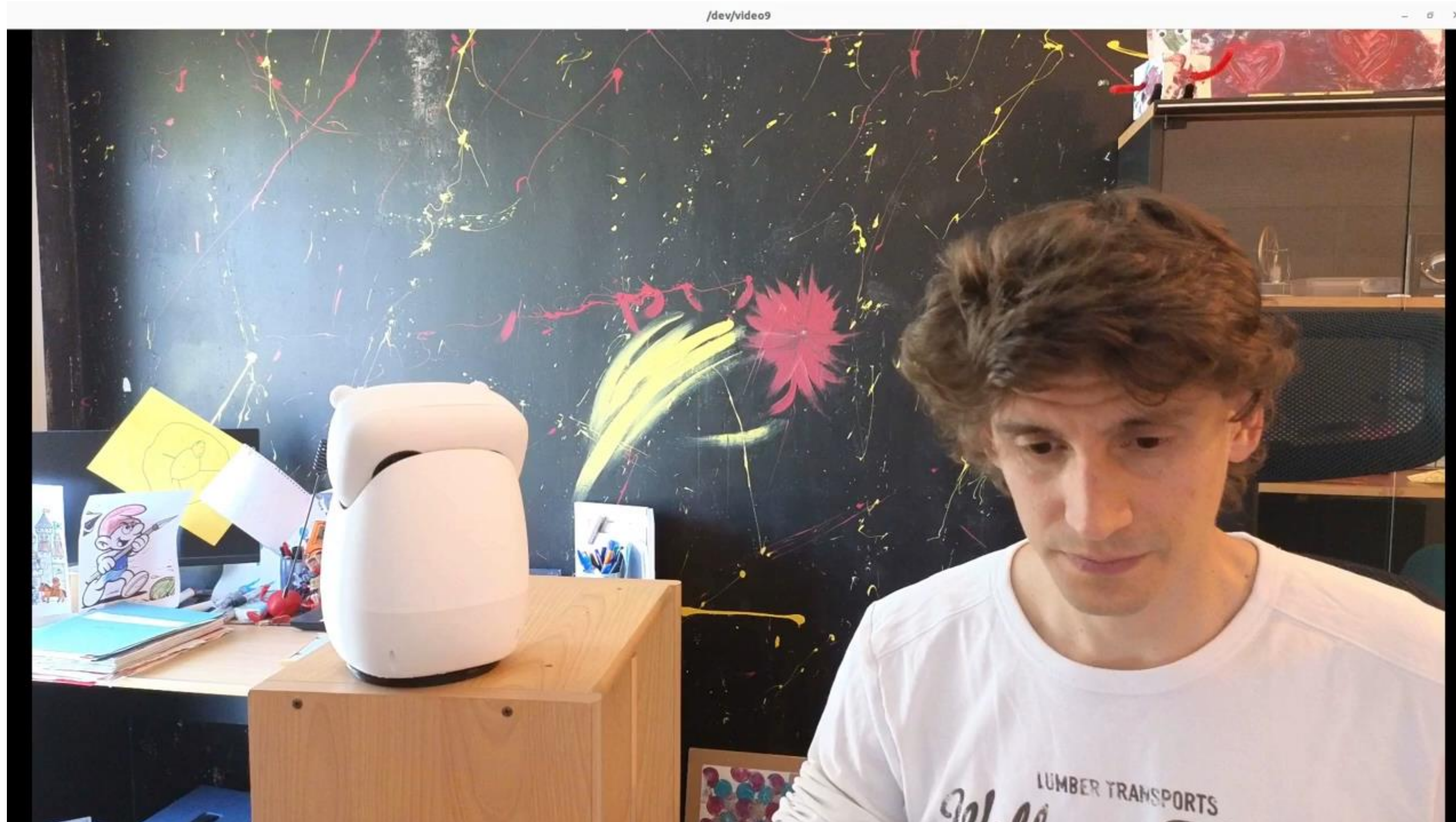


Cleaning Robot

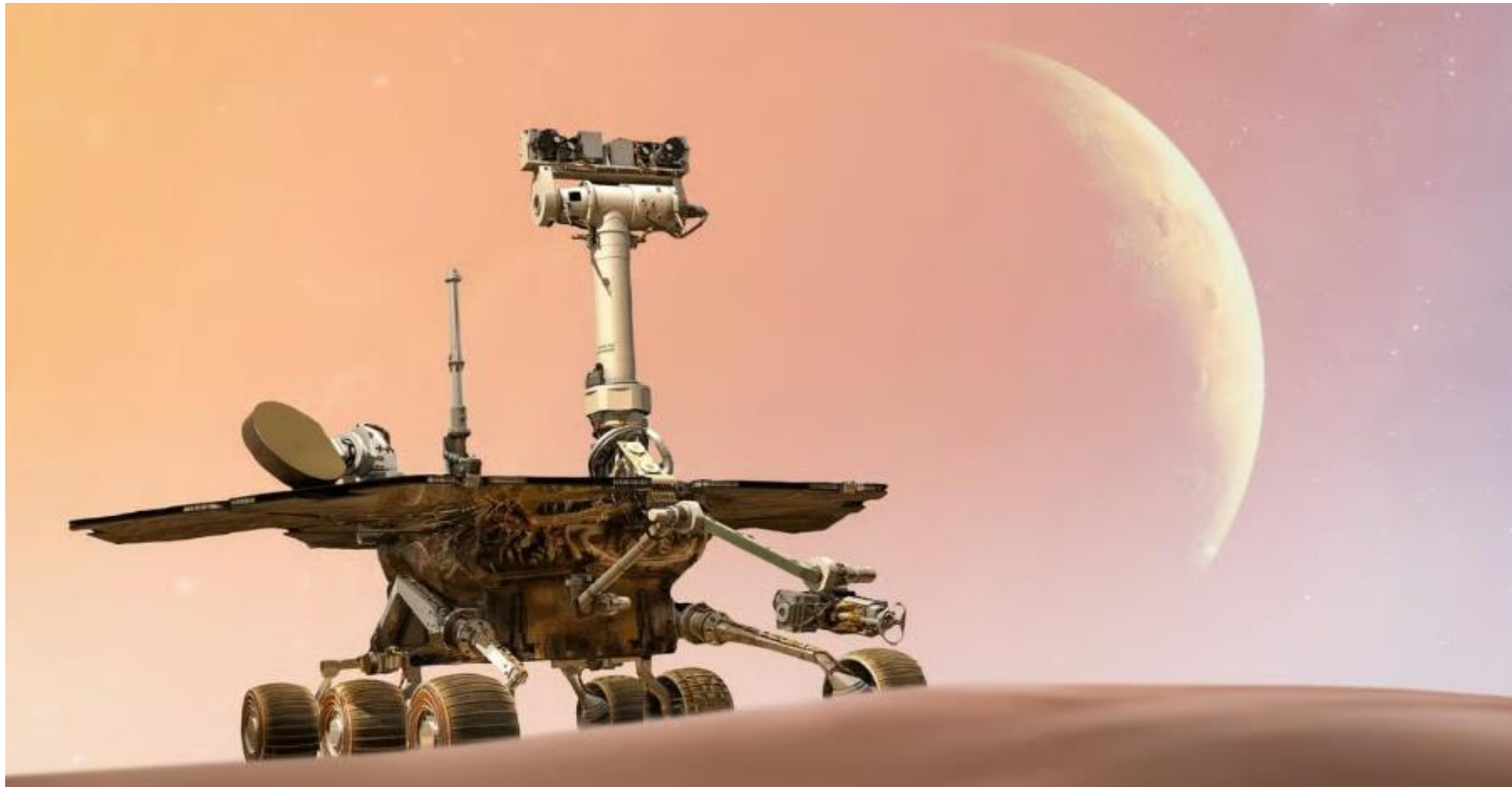


Companion robots

Home & Daily Life



Exploration & Hazardous Environments



Education & Entertainment



LEGO® MINDSTORMS

What will we study?

- **Mechanical**

- Motors
- Gears
- Mechanism design

- **Electrical**

- Sensors
- Microcontrollers
- Motor drivers

- **Software**

- Flow charts
- Programming algorithms
- Debugging and testing

- **Multi-body Kinematics**

- Forward kinematics
- Inverse kinematics

- **Robotic Operating System (ROS)**

- Basic components of ROS
- Programming with ROS

What will we do?

- Design and build robots
- Use sensors and actuators
 - Wire circuits
 - Characterize performance
- Read reference material
- Do projects
 - Write code
 - Develop problem solving skills
 - Develop troubleshooting skills
 - Navigate mazes
 - Get creative

How will your grade be determined?

Grading

There will be:

- Homework (individual effort)
- Exams: one midterm (individual effort)
- Projects/labs (group effort, 60% of grade is based on completion of the lab activities and 40% on the report)
- Robot of the Week presentation (presented in lecture, group presentation)

Activity	% of grade
Homework	20
Midterm Exam	10
Labs 1-4	40
Final Project	20
Robot of the Week Presentation (1)	5
Attendance and Participation	5

Arduino C programming is needed

*ROTW and Projects are team scores, all others are individual

Robot of the week presentation

- Introduce any robot you want to present.

Lab projects

- Motor Performance Measurements
- Open-Loop Control
- Position Control
- Velocity Control
- Maze Solving
- Open design projects – overall demonstration of learning

Office Hours

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